FAROOQ HOSPITAL

(Bahria Enclave Islamabad)

Hospital Road, 3/5, Sector Commercial-C, Bahria Enclave Housing Scheme,
Zone-IV-C Islamabad. Tel: 051 - 2245241 - 2245242

Farooq Hospital
(Westwood Branch)
262-263, Westwood Colony,
Thokar Niaz Baig, Lahore.

Farooq Hospital
(Allama Iqbal Town Branch)
2-3 Asif Block,
Main Boulevard,
Allama Iqbal Town, Lahore.

Farooq Hospital
(Islamabad Branch)
Adjacent Bahria Golf City,
Murree Expressway,
Rawalpindi.

Farooq Hospital
(DHA Branch)
P Block, DHA,
Main Ghazi Road,
Rehman Gardens
Near Bhatta Chowk, Lahore.

Akhtar Saeed Trust Hospital
258-F, EME D.H.A. Sector,
Multan Road, Lahore.

Akhtar Saeed Medical & Dental College
Tulip Block Sector C,
Bahria Town, Lahore.

Akhtar Saeed Medical & Dental College
(Islamabad Campus)
Bahria Golf City,
Murree Expressway,
Rawalpindi.

Akhtar Saeed Group of Pharmacy Colleges
(Lahore & Islamabad)

College of Allied Health Sciences *(Lahore)*

Akhtar Saeed College of Rehabilitation Sciences
(Lahore)

College of Nursing
(Lahore)

Avenue Mall
P Block, DHA,
Main Ghazi Road,
Rehman Gardens
Near Bhatta Chowk,
Lahore.

Reference # FH/BE/01/25

Date: 11th April, 2025

Experience Certificate

It is certified that **Muhammad Khaleeq Saqi S/O Muhammad Saghir Saqi** has worked in **Farooq Hospital** Bahria Enclave Branch as **Laboratory Technologist** from **03rd September, 2024** to **15th March, 2025.**

During his employment at **Farooq hospital**, I found him regular, dutiful and efficient. He worked hard and gained good experience.

His performance & relation with other staff was excellent.

Dr. Saad Bin Haroon
Chief Administrative Officer &
Medical Superintendent
Farooq Hospitals, Islamabad
Email: fh.isd@farooghospitals.com
Tel: (+92 51) 2243322-3

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IBADAT INTERNATIONAL UNIVERSITY
ISLAMABAD

IIUI/FAHBS/MLT/01/30.11.2025

TO WHOM IT MAY CONCERN

It is certified that **Mr. Muhammad Khaleeq Saqi S/O Muhammad Saghir Saqi** CNIC No: 81203-9733909-7 worked from17th February 2025 to 30th November 2025 in the Department of Medical Lab Technology, discipline of the Allied Health and Biological Sciences at the Ibadat International University, Islamabad as a Lecturer.

I found him honest, obedient and capable person and wish him a bright career.

30.11.2025

Head of the Department
Medical Laboratory Technology
Ibadat International University, Islamabad
Email: ammar.ahmed@mlt.iiui.edu.pk
Contact number: +923015319706

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COMSATS UNIVERSITY ISLAMABAD (CUI)

Park Road, Tarlai Kalan, Islamabad. Tel: +92-51-9049-6107, Cell: +92-3215330626
Email: : maleeha.azam@comsats.edu.pk URL://www.comsats.edu.pk

Department of Biosciences

Friday, Oct 11, 2024

To Whom It May Concern

I am writing to formally acknowledge **Muhammad Khaleeq Saqi** for his significant contributions as a **Research Assistant** on the project titled **"Mitochondrial DNA Analysis in Strabismus Patients"** at the Translational Genomic Laboratory (TGL) of COMSATS University Islamabad for one year. Throughout his time with us, Khaleeq showcased an outstanding ability in both molecular biology techniques and laboratory management, which greatly enhanced the efficiency and productivity of our lab.

Khaleeq possesses hands-on experience with a range of essential molecular biology techniques. He has skillfully performed DNA extraction, PCR (Polymerase Chain Reaction), gel electrophoresis, Next Generation Sequencing, and other related protocols crucial to our research goals. His capability to execute these techniques with accuracy and dependability has been vital to the progress of our projects. In addition to his wet lab skills, he also has expertise in bioinformatics. During his tenure, he conducted labs and classes for undergraduate students at this university. Beyond his technical abilities, Khaleeq has effectively overseen various aspects of laboratory operations. His duties included managing lab inventory, ordering supplies, and ensuring that all necessary reagents and equipment were available for ongoing research. His meticulous attention to detail and organizational skills have been instrumental in minimizing disruptions and ensuring that our lab remains well-stocked and functional at all times. Khaleeq's commitment to his work, along with his technical knowledge and management skills, has made him an invaluable asset to our team. I wholeheartedly recommend him for any future opportunities and am confident that he will thrive in any research setting.

Please feel free to reach out to me if you need any additional information.

Maleeha Azam
Associate Professor
Translational Genomics Laboratory
Department of Biosciences
COMSATS University Islamabad, Pakistan
Cell: +92 3215330626 | Office: +92 51 90496107
Email: maleeha.azam@comsats.edu.pk

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Russian Scholarship Project
"Global Universities" Association

DIPLOMA OF THE WINNER

awarded to

Saqi Muhammad Khaleeq

in recognition of success in

Biology and Biotechnology

in the Doctoral track

Moscow, Russian Federation
21.02.2025

№ 2024-DT-BIO-033

MUHAMMAD KHALEEQ SAQI

Email ✉: |Khaleeqsaqi15@gmail.com|ORCID:0009-0002-5419-6013| Adress: Islamabad, Pakistan

|Phone: +923419397785|Nationality: Pakistani

PROFESSIONAL SUMMARY & Current Position

I am doing my PhD in Genetics from Southern Federal University Rostov on don, Russia and am a motivated researcher with a strong background in human genetics, molecular biology, and transcriptomics. I completed my master's research on *"Mitochondrial DNA Analysis in Strabismus Patients,"* representing a pioneering study exploring the genetic basis of this multifactorial ocular disorder, and I am currently involved in transcriptome (RNA-seq) analysis of strabismus, focusing on differential gene expression and pathway analysis. I possess hands-on experience in DNA/RNA extraction, PCR-based assays, NGS, bioinformatics analysis, and variant interpretation, with familiarity in CRISPR/Cas9 functional genomics. With over three years of laboratory and research experience and my current role as a Lecturer at Ibadat International University Islamabad, I am committed to advancing research on genotype–phenotype correlations, gene regulation, and molecular mechanisms underlying ophthalmic and neurodevelopmental disorders.

KEY ACHIEVEMENTS

- Appointed as a Lecturer at Ibadat International University, Islamabad, Pakistan.
- Led the first baseline study named Mitochondrial DNA Analysis of Strabismus Patients in Pakistan.
- Published my first co-author research paper on Hepatitis C Virus Frequency in September 2024, which is my first contribution to the scientific community.
- First in my family generation to have completed a master's degree and am pursuing a PhD in the relevant area.

RESEARCH SKILLS

- **Molecular Biology Expert:** DNA and RNA extraction, Gel electrophoresis, PCR, RT/qPCR, ARMS PCR, RNA Seq, WES
- **Genomic and Bioinformatics Analysis:** Next Generation Sequencing, Data analysis (GraphPad Prism, SPSS, R Software, Python), Phylogenetic analysis (MEGA Software), BLAST (Basic Local Alignment Search Tool) for sequence similarity, ClustalW or MUSCLE for sequence alignment, Primer designing. Flowchart creation (Biorenders.com).
- **Geospatial Mapping:** GIS mapping (ArcGIS PRO, QGIS), Maxent modeling.
- **Biochemical and Serological Assays:** Enzymatic Assays, serological assays (ELISA, Western blotting), Flow Cytometry, Cell Culture, Immunohistochemistry, Microscopy and Immunofluorescence
- **Field Work and Sampling Skills:** Blood Sample Collection, Setting sample selection criteria.
- **Research administration and project management skills:** Grant writing, Proposal Development, identifying potential funding sources and aligning project goals with their interests, Budget Planning, and Research and Analysis.

RESEARCH PUBLICATIONS

1. **Hepatitis C Virus Frequency in Hemodialysis Patients**. Pakistan Armed Forces Medical Journal, 74(3), 861–864. Rathore, M. A., **Saqi, M. K.,** Ghani, E., Niazi, S. K., ahmed, faraz, & Hammad, H. (2024). https://doi.org/10.51253/ pafmj.v74i3.10297
2. **Genotype association of VEGF polymorphism rs2010963 with Strabismus in a Pakistani cohort**. bioRxiv (2025). Authors: Khalid, Muhammad., Amin, Noor-ul-ain., Nehal, Conain., **Saqi, Muhammad Khaleeq**., Woo, Kwang.,Jamil, Muhammad Usman., Azam, Maleeha. DO - 10.1101/2025.10.23.25338643

EDUCATION

COMSATS University Islamabad **Sept. 2022 – June. 2024** *Master of Science in Biochemistry and Molecular biology (Focused on Genetics)* *Islamabad, Pakistan* ***CGPA:*** 3.75/4.0

Thesis: *"Mitochondrial DNA Analysis in Strabismus Patients"*

Supervisor: Dr. Maleeha Azam (Associate Professor)

National University of Medical Sciences Islamabad (NUMS) **Jan. 2018 – April 2022**

Bachelor of Science in Medical Lab Technology (Focused on Infectious Disease) *Islamabad, Pakistan* ***FYP Project:*** *"Hepatitis C Virus Frequency in Hemodialysis Patients"*

CGPA: 3.62/4.0

Supervisor: Lt Col Saif Ullah Khan Niazi (Consultant Virologist)

RELEVANT COURSEWORK

Advances in biochemistry, Advances in Molecular biology, Genetics, Synopsis and Research Methodology, Biostatistics

PROFESSIONAL EXPERIENCE

Ibadat International University, Islamabad **Feb. 2025 – 01 December**

Lecturer, Department of Medical Lab Technology

- Teaching Basic & Advanced Biochemistry, Genetics, and Advanced Molecular biology to undergraduates.
- Supervised multiple bachelor's research projects on prevalence, risk factors and genetics of Strabismus and other Neurodevelopmental disorders.

COMSATS University Islamabad **June. 2023 – July 2024**

Research Assistant, Department of Biosciences *Islamabad, Pakistan*

- Assisted supervisor in crafting student synopses and developing thesis.
- Taught data analysis and visualization techniques to undergraduate and graduate students.
- Provided support in the preparation of research manuscripts and academic papers.

*Professional Mentorship, COMSATS University Islamabad***.**

- Mentored and supervised 3 undergraduate and 4 master's students during a semester.
- Facilitated and provide training to students on blood sample collection.
- Conducted training sessions on data analysis tools (e.g., GraphPad Prism, SPSS, R Software) and best practices for interpreting and presenting research findings.
- Oversaw laboratory chemical inventory and logistics, supporting student practical work.
- Develop expertise in molecular techniques DNA and RNA extraction, Gel electrophoresis, PCR, RT/qPCR, ARMS PCR, RNA Seq, WES
- Ability to develop, optimize, and troubleshoot experimental protocols
- Liaised with suppliers and ensured efficient reagent availability for academic activities.

Farooq Hospital Bahria Enclave Islamabad, Pakistan **Sept 03, 2024 - March 15, 2025**

Medical Lab Technologist *Islamabad, Pakistan*

- Supported cellular and molecular workflows relevant to disease mechanism studies, including cell culture and microscopy-based cellular analysis and quality-controlled biochemical assays
- Worked on Flow cytometry, CLIA, CMIA and other biochemical assays

Maryam Memorial Hospital Bahria Town, Rawalpindi **Sept 14, 2020- Sept 15, 2022**

Medical Lab Technologist *Islamabad, Pakistan*

- Applied molecular biology techniques such as DNA extraction and PCR to analyze genetic samples, enhancing lab efficiency and accuracy.
- P. Film examination for diagnosis of Anemia and Malarial parasites and implementation of Quality control.

Armed Forces Institute of Pathology, Rawalpindi **Feb. 2019 – March 2022**

Internie *Rawalpindi, Pakistan*

For three years of training at AFIP during undergraduate studies, I worked independently in all departments which includes

- Hematology, Microbiology, Histopathology, Chemical Pathology, Immunology, Molecular biology and Virology
- Learned different research techniques like PCR, ELISA, western blotting, Flow cytometry, Cell culture, IHC and Immunofluorescence. Exposed to working in Biosafety Level 3 Labs.

RESEARCH PROJECTS

COMSATS University Islamabad | *Graduate Researcher* **Sept. 2022 – July. 2024**

Project: " *Mitochondrial DNA Analysis in Strabismus Patients*"
"Transcriptome Analysis of Strabismus Patients"
"DNA Methylation Pattern in Strabismus Cases and Controls"

- Conducted the first study in Pakistan on Mitochondrial DNA analysis of Strabismus Patients
- Applied the whole exome sequencing technique to detect novel mitochondrial DNA variants.
- Reported several differentially expressed genes and differentially methylation regions which are not reported before.
- Research Papers are in process as we are combining all these analyses, and we used the same sample, and it will be a high impact factor paper.

National University of Medical Sciences, Islamabad (NUMS) | *Undergraduate Researcher* **Feb. 2021 – March 2022**

Project: *"Hepatitis C virus Frequency in Hemodialysis Patients"*.

- Explored the prevalence and distribution of HCV in Pakistani Population.
- Investigated the potential risk factors associated with HCV such as blood transfusion, IV drug usage, Surgery and usage of multiple dialysis centres.

COURSES AND CERTIFICATIONS

- Data Analysis in R Programming (One Month) **Sep 6, 2024- Oct 06, 2024**
- The Systems Approach to ArcGIS: An Introduction **August 10, 2024**
- Certified Peer Reviewer Course **Monday 12 April, 2024**
- Unlocking Grant Success with Funding Institutional (43 minutes) **Monday 15 March, 2024**
- Funding (Researchers academy) **Tuesday 13 June, 2024**
- Funding Hacks for Researchers (61 minutes) **Monday 08 Feb, 2024**
- SPSS DATA ANALYSIS (Two weeks) **Dec 15, 2023-Dec 30, 2023**

HONORS & AWARDS

- **Certificate of Professional Development**: Infections and Injunctions: Update on diagnosis and management of Infectious Uveitis, Cybersight. **22 December, 2025**
- **Certificate Of Recognition:** Awarded On participation in Fully Funded International Genomics and Bioinformatics Bootcamp (Genomac Institute Inc USA). **Nov 21,2024**
- **AJK State Talent Scholarship Award 2023:** Granted from the Higher Education
- **Need Base Scholarship Award:** Awarded to students during intermediate and continues until graduation based on satisfactory academic performance. **July. 2019 – March. 2022**
- Department of AJK Government during Master's studies on achieving high grade and maintenance them throughout the degree program. **Oct. 2019 – Aug. 2021**

POSTER PRESENTATIONS, CONFERENCES & WORKSHOPS

- **COMSTECH Hybrid Symposium** "Preventing Inherited Disorders in Consanguineous Population" A Framework for Action, **COMSTECH**. 02 August 2025.
- **International Webinar Series for Life Scientists, Genomac Institute Inc. USA.** 13 December, 2024.
- **Basic of Pythons**, **Cambridge International Qualifications, UK.** September 24, 2024.
- **ICESCO'S 1st International Next Generation Sequencing Data Analysis Workshop**
  ICESCO / INTERGEN / COMSATS. 28 September, 2023.
- How to Develop New Multiplex SARS-COV-2 Variant Neutralizing Antobody Assay in a Day
  BIO RAD. 12 December, 2021
- Poster presentations on "**Dengue Infection in Pakistan and Its Controlling Strategies**" at the International Conference and Workshop QAU Islamabad, Pakistan. 15 December, 2020.

LANGUAGES

Urdu C2 & English C1 Proficiency

PROFESSIONAL ASSOCIATIONS

- World Society of Pediatric Ophthalmology and Strabismus (WSPOS)
- Childhood Glaucoma Research Network (CGRN)
- Pakistan Society of Medical Virology

REFERENCE

Reference I:
Dr. Maleeha Azam (Master's Supervisor)
Supervisor)
Associate Professor
Department of Biosciences
COMSATS University, Islamabad
Email: maleeha.azam@comsats.edu.pk
Phone: +92 -3215330626

Reference II:
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Associate Professor
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COMSTAS University Islamabad, Pakistan
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Tel: +92-3005347366